

Yiyuan (Steve) Wufeng

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Computer Science Graduate · Seeking Full-Time Software Engineer Position · U.S. Citizen

EDUCATION

Rochester Institute of Technology

Bachelor of Science in Computer Science, Minor: Mathematics

Rochester, NY

Aug 2021 – May 2026

Cumulative GPA: 3.85/4.00

Relevant Coursework: Large Language Models, Artificial Intelligence, Computer Graphics, Data Management, VR/AR Interactive Music, Simulation & Randomness, Ethics in Leadership

EXPERIENCE

Full-Stack Developer (Personal Project)

E-Commerce

Jan 2026 – Present

Spokane, WA

- Architected a production-ready full-stack e-commerce platform with Next.js 14, TypeScript, and PostgreSQL (Prisma ORM)
- Containerized the application using Docker with multi-stage builds and deployed on Railway with automated CI/CD via GitHub Actions
- Integrated Cloudflare R2 (S3-compatible) for product image and payment receipt storage with secure upload flows
- Implemented a complete order lifecycle (PENDING → CONFIRMED → READY_FOR_PICKUP → PICKED_UP / REJECTED) with admin receipt review panel
- Built RESTful API layer handling authentication (NextAuth.js), cart management, product CRUD, and analytics endpoints
- Wrote unit and integration tests using Jest; configured Docker Compose for local PostgreSQL development

Embedded Software Engineer

Waygate Technologies (Baker Hughes)

Jul 2025 – Dec 2025

Skaneateles, NY

- Engineered automated data processing and CI/CD pipelines to track production quality metrics, reducing manual inspection overhead
- Trained OCRNet with NVIDIA TAO to detect defects and low-quality X-ray outputs in industrial inspection workflows
- Built and deployed computer vision models into production testing environments using DirectML and C++ inference pipelines
- Developed 4 automation test cases for pipeline validation, improving regression coverage for build releases
- Processed and analyzed large-scale inspection data to identify failure patterns and optimize model accuracy

Jr. Software Developer

Tive Technology

Jun 2024 – Sep 2024

Remote

- Developed 4 mobile app features using Flutter and Android Studio
- Designed 6 clean, responsive UI pages following Material Design guidelines embedded in the production app
- Built backend services with Firebase including Firestore data modeling, Authentication, and Security Rules
- Developed and maintained RESTful APIs with SpringBoot for server-side business logic

Research Assistant

RIT CAIR Lab

Sep 2023 – May 2024

Rochester, NY

- Implemented a real-time transcription web service integrating Google Cloud Speech-to-Text with prosody and emotion

analysis to assist deaf and hard-of-hearing users

- Collected and analyzed experiment data comparing traditional vs. affective captioning systems for accessibility research
- Analyzed eye-tracker data using Python and contributed findings to a paper presented at the ASSETS conference
- Built the frontend interface using React and Vue.js for live captioning visualization

Research Assistant

RIT Global Cybersecurity Institute

Sep 2022 – Jul 2023

Rochester, NY

- Performed statistical analysis and processing on MITRE threat alerts to identify security patterns
- Set up space cyber-physical simulations with NS-3 for satellite communication and motion modeling
- Analyzed real-time satellite motion and communication systems using MATLAB and C++

Research Intern

Tufts University

May 2022 – Jul 2022

Medford, MA

- Optimized properties of novel polymers using gradient descent algorithms in MATLAB and C++
- Applied object-oriented class structure and documentation practices to the Morpho-Lang compiler project
- Solved compiler and modular design problems for Morpho-Lang, an open-source scientific computing language
- Assembled a research poster presented to material science experts

PROJECTS

NeuralCube – Open Source Autonomous Robot

Collaborator (with Chanel F. Cheng)

2024 – Present

Remote

- Collaborated on an open-source autonomous robot orchestrated by multi-agent LLMs for natural interaction and transparent reasoning
- Deployed LLM inference serving using vLLM and SGLang for real-time model serving on the robot
- Contributed to the digital twin system that maintains a continuously updated self-model for simulation-based reasoning
- Integrated computer vision pipelines for visual perception and navigation in real-world environments
- Tech stack: Python, vLLM, SGLang, Multi-Agent Systems, Computer Vision, Robotics

4G LTE Unlimited Distance Drone

Personal Project

Aug 2023 – Dec 2024

Rochester, NY

- Built a fully programmable 4G LTE drone using Raspberry Pi with internet-based communication for beyond-visual-line-of-sight operation
- Implemented server communication for real-time image segmentation and autonomous navigation commands
- Integrated PixHawk flight controller with ArduPilot for stabilized flight and Sixfab 4G LTE cellular hat for connectivity
- Tech stack: Python, Raspberry Pi, ArduPilot, PixHawk, Computer Vision, IoT

KittyFan – Persistence of Vision Display

Personal Project

2022

Rochester, NY

- Created a persistence of vision (POV) display using programmable LED strips and 3D-printed rotating hardware
- Implemented mathematical visual logic (Calculus II) for image rendering on a spinning display
- Tech stack: Python, Raspberry Pi, 3D Printing, Electronics

TECHNICAL SKILLS

Languages: Python, Java, C#, C/C++, TypeScript, Dart, MATLAB, SQL
Web & Mobile: Next.js, React, Vue.js, Angular, Flutter, HTML/CSS, Node.js
Cloud&DevOps: Docker, GitHub Actions (CI/CD), Railway, Cloudflare R2, Firebase, Google Cloud Platform, Azure DevOps
AI/ML: NVIDIA TAO, OCRNet, vLLM, SGLang, Large Language Models, Computer Vision, DirectML
Databases: PostgreSQL, Prisma ORM, Firestore, MySQL
Tools: Git, Visual Studio Code, Vim, Android Studio, Figma, Proxmox, Virtual Machines
Hardware: Raspberry Pi, ESP32, 3D Printing (OnShape), PixHawk, ArduPilot, Sixfab LTE

LEADERSHIP & COMMUNITY

Computer Science House (CSH) *Aug 2021 – Present*
Active member of RIT's CS-focused living-learning community. Hosted 5 technical seminars and workshops on DevOps, Linux administration, and software development to mentor peers and spread knowledge across the community.

RIT Artificial Intelligence Club *Aug 2022 – Present*
Member of AI enthusiast community. Explore active research areas in AI/ML with fellow students and collaborate on group projects. Participated in discussions on large language models, computer vision, and autonomous systems.